

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A 2-Part, Phase 1b Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of BMS-986454 in Participants With Rheumatoid Arthritis **Short Title/ Acronym:** A Study to Evaluate the Safety, Tolerability, and Drug Levels of BMS-986454 in Participants With Rheumatoid Arthritis **Protocol Number:** IM055-1014 **NCT Number:** NCT07171983 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** BMS-986454 **Phase of Development:** Phase 1b **Medical Monitor:** BMS Clinical Trials Contact Center

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, and drug levels of BMS-986454 in participants with Rheumatoid Arthritis. **Secondary Objectives:** To evaluate the pharmacokinetics (PK) and pharmacodynamics (PD) of BMS-986454, specifically assessing maximum observed concentration (Cmax), concentration at the end of a dosing interval (Ctau), trough observed plasma concentration (Ctrough), time of maximum observed concentration (Tmax), and area under the curve (AUC) over the dosing interval.

2.2 Background & Rationale To address the ongoing medical necessity for advanced therapeutic options in Rheumatoid Arthritis (RA). This study evaluates the novel investigational agent BMS-986454 in patients who have been incomplete responders to prior methotrexate (MTX) treatment, assessing its foundational safety and PK/PD profiles to guide further clinical development.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Double Masking) **Randomization:** Randomized, Parallel Assignment

3.2 Planned Enrollment & Duration Target \$N\$: 46 participants **Number of Sites:** Multicenter **Estimated Study Start:** 28-Feb-2026 **Estimated Primary Completion:** 28-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants with Rheumatoid Arthritis must have disease onset after age 18. 2. Participants who meet the 2010 American College of Rheumatology/European League Against Rheumatism (ACR/EULAR) classification criteria for RA. 3. Participants must have evidence of swelling in at least 1 joint of the hand or wrist by clinical examination at screening and Day -1. 4. Participants must have been an incomplete responder to prior methotrexate (MTX) treatment.

4.2 Exclusion Criteria 1. Participants must not have any significant medical condition, with the exception of Rheumatoid Arthritis (including but not limited to, neurological, GI, renal, hepatic, cardiovascular, psychological, pulmonary, metabolic, endocrine, hematological, allergic disease, drug allergies, or other major disorders) that, in the Investigator's judgment, will substantially increase the risk to the participant. 2. Participants must not have any condition aside from RA that confounds the ability to interpret data from the study. 3. Participants must not have severe Rheumatoid Arthritis as assessed by Disease Activity Score 28 c-reactive protein (DAS28-CRP) at screening or Day -1.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Specified doses of BMS-986454 or matching placebo administered on specified study days across the 2 parts of the Phase 1b trial. **Route of Administration:** Investigational formulation per protocol. **Duration of Treatment:** Up to 6 months.

5.2 Concomitant & Prohibited Therapy Permitted: Prior use of methotrexate (MTX) as base standard-of-care history; specific permitted concomitant therapies per protocol guidelines. **Prohibited:** Therapies that could confound PK/PD assessments or substantially increase safety risk to the participant.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -28 to -1 Informed Consent, Medical History, Joint Examination, DAS28-CRP
Baseline	Day -1 to Day 1	Randomization, First Dose, Pre-dose PK Sampling	
Interim	Over 6 months	Safety Labs, Efficacy Assessment, Cmax/Ctrough PK Blood Draws	
End of Study	Up to 6 months	Final Vital Signs, Exit Interview, IP Accountability, Final PK/PD	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Assessment of the safety and tolerability profile of BMS-986454, primarily measured via the incidence of treatment-emergent Adverse Events (AEs), Serious Adverse Events (SAEs), and clinically significant changes in vital signs or laboratory values. **Measurement Method:** Clinical evaluation, laboratory monitoring, and medical review throughout the study duration.

7.2 Secondary & Exploratory Endpoints * Maximum observed concentration (Cmax) up to 6 months. * Concentration at the end of a dosing interval (Ctau) up to 6 months. * Trough observed plasma concentration (Ctrough) up to 6 months. * Time of maximum observed concentration (Tmax) up to 6 months. * Area under the curve over the dosing interval (AUC) up to 6 months.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous reporting and collection via Investigator examinations and patient logs.

Serious Adverse Events (SAEs): Must be reported to the Sponsor (Bristol-Myers Squibb) within 24 hours of site awareness.

Data Monitoring Committee (DMC): Utilized to monitor early Phase 1b safety thresholds and dosing safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all safety parameters (all participants receiving ≥ 1 dose); PK/PD Analysis Set for drug level endpoint calculations. **Sample Size Justification:** A sample size of $N=46$ is planned to adequately characterize the preliminary safety, tolerability, and PK/PD profile of BMS-986454 in human participants. **Handling of Missing Data:** PK/PD parameters calculated based on available data points; missing safety data handled according to institutional phase 1 statistical protocols.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Sponsor-directed clinical monitoring visits to verify adherence to protocol, source document accuracy, and Investigational Product (IP) accountability. **Audit Trail:** Robust Electronic Data Capture (EDC) locking, audit trails for PK/PD samples, and data clarification forms.

11. Study Identifiers & Governance

Primary ID: IM055-1014 **Registry Number:** NCT07171983 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** A Study to Evaluate the Safety, Tolerability, and Drug Levels of BMS-986454 in Participants With Rheumatoid Arthritis **Official Regulatory Title:** A 2-Part, Phase 1b Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of BMS-986454 in Participants With Rheumatoid Arthritis

12. Clinical Framework

Primary Condition: Rheumatoid Arthritis (RA) **Diagnosis Threshold:** 2010 ACR/EULAR classification criteria with joint swelling. **Medical Methodology:** Assessment of Investigational Therapy (BMS-986454 vs. Placebo). **Optimization Target:** Confirming tolerability, establishing safety profiles, and capturing comprehensive drug PK/PD levels (Cmax, Tmax, AUC).

13. Study Procedures & Methodology

Management Pathway: Targeted Phase 1b evaluation for MTX-IR Rheumatoid Arthritis. **Education Protocol (HCP):** Sponsor training on blinded IP administration, complex PK blood draw schedules, and AE reporting. **Education Protocol (Patient):** Thorough informed consent process covering early-phase experimental risks, expected visit schedules, and AE self-reporting. **Quality Audit Mechanism:** Routine oversight of site operations, strict DAS28-CRP evaluation consistency, and lab sampling accuracy tracking.

14. Observation & Follow-Up Schedule

Total Duration: Up to 6 months. **Onsite Visit Cadence:** Frequent visits centered around the 2-Part dosing schedule to map accurate drug levels. **Remote Monitoring Frequency:** Supplemental to physical site audits. **Checklist Review Interval:** Regular interval reviews for ongoing signs of joint swelling and treatment tolerability.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				